Q1 Solutions

Import data set to R assigning the type of each variable correctly

```
air_data <- read.csv("./data/airplane_price_dataset.csv", sep=",",
stringsAsFactors=TRUE)
air_data <- air_data |>
  rename(
    FC = FuelConsumption.L.h.,
    HM = HourlyMaintenance...,
    Price = Price...
  )
air_data$EngineType <- as.factor(air_data$EngineType)
air_data$NumberofEngines <- as.factor(air_data$NumberofEngines) # only 2
different amount of engines
air_data$ProductionYear <- as.factor(air_data$ProductionYear)

str(air_data)
```

After analyzing the data we decided to:

- Rename long name variables for better handling.
- Convert to factors `EngineType`, `NumberofEngines`, and `ProductionYear` as they have little variation.

Summarize the variables Price, Fuel Consumption first for the all data and then for the groups of Engine Type

Total summary

```
> summary(air_data$Price)
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
   145815  14096814  83921914 198833650 384323881 978213229
> summary(air_data$FC)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   2.00    5.95    9.82   12.08   13.47   49.97
```

Figure 1.1

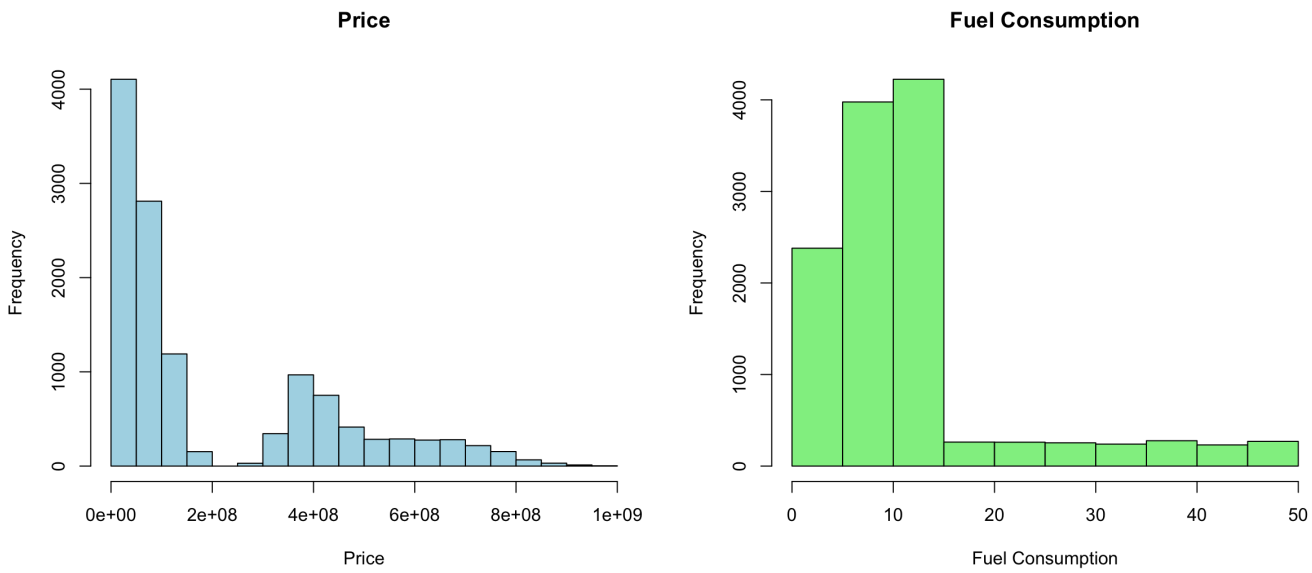

Figure 1.2

Summary by Model

	EngineType	Price_Mean	Price_SD	Price_Median	Price_Min	Price_Max	FC_Mean	FC_SD	FC_Median	FC_Min	FC_Max	n
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>
1	Piston	279405.	77369.	251474.	145815.	561279.	30.1	11.6	29.9	10.0	50.0	2039
2	Turbofan	237995200.	231292861.	108876395.	8150222.	978213229.	8.52	3.75	8.53	2	15	10338

Figure 2.1

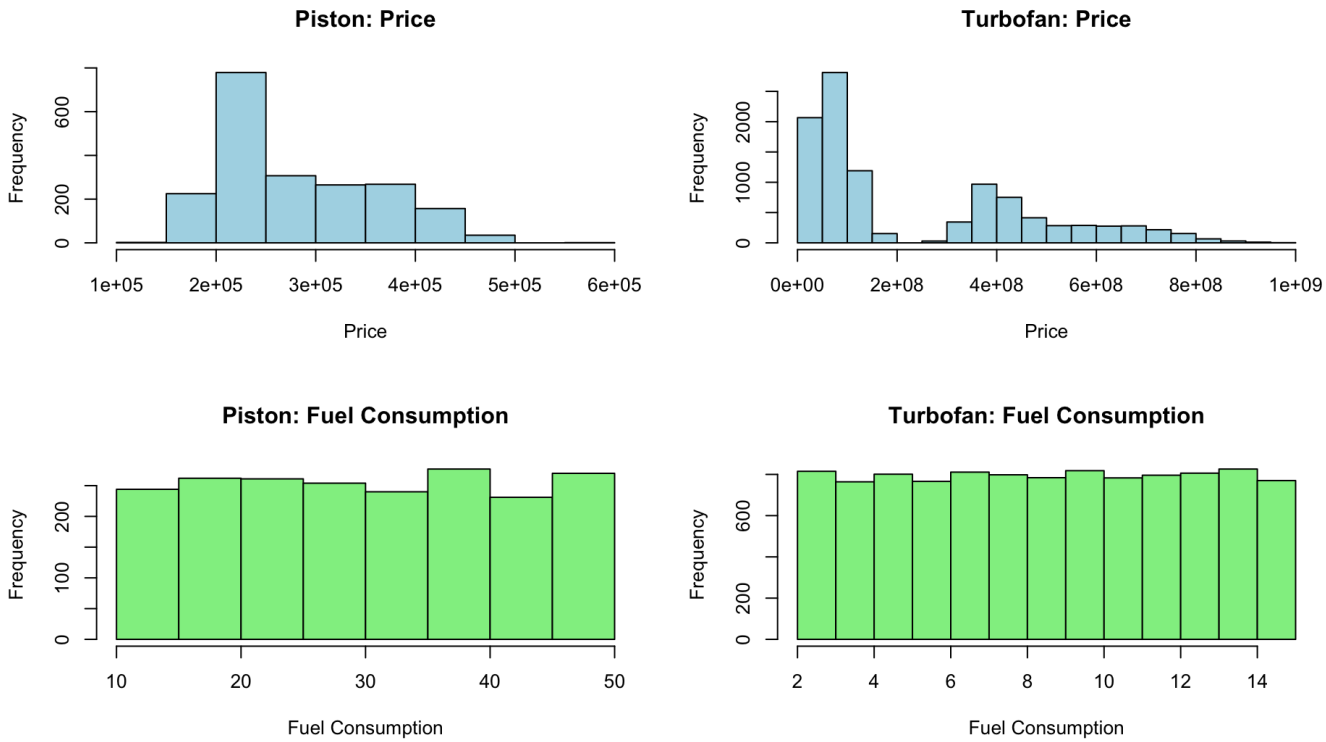

Figure 2.2

Test whether Fuel Consumption is affected from the Engine Type of the plane. Check the assumptions and visualize the relationship between these two characteristics

Comparing distributions

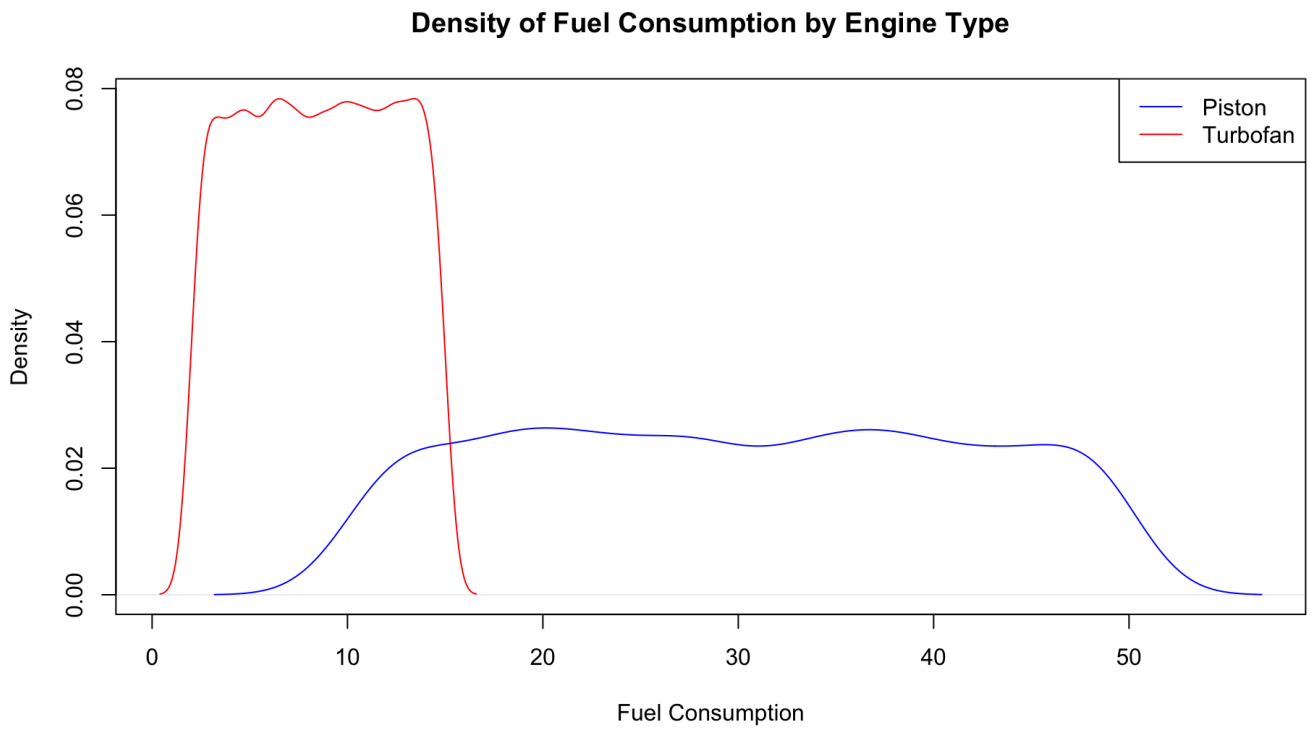

Figure 03

Boxplot is best for comparing a continuous variable across categories

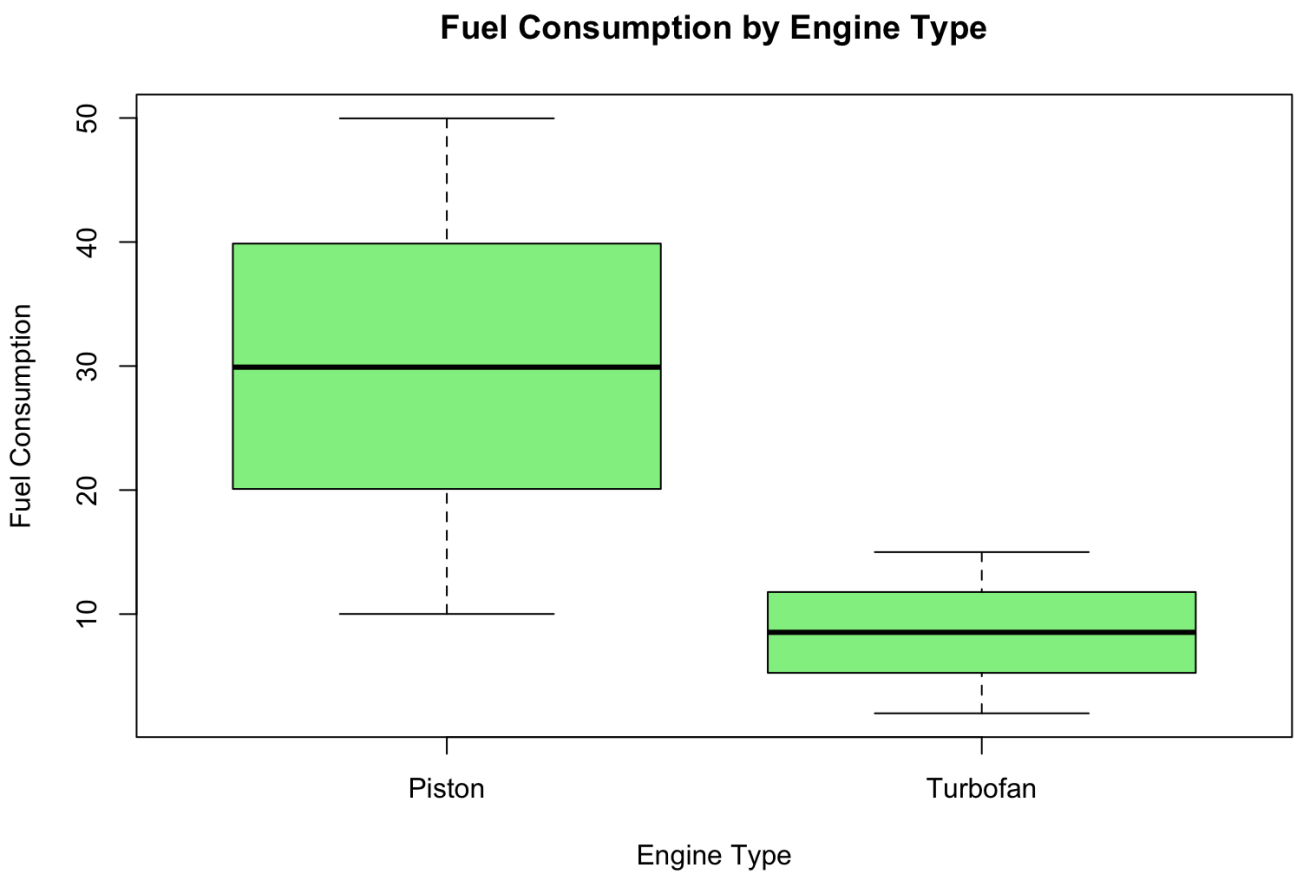

Figure 04

```
        Welch Two Sample t-test

data:  FC by EngineType
t = 83.347, df = 2123.1, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Piston and group Turbofan is not equal to 0
95 percent confidence interval:
 21.09784 22.11459
sample estimates:
  mean in group Piston mean in group Turbofan
            30.122398               8.516185
```

Figure 05

Conclusion

Since the p-value is < 0.05, we reject the null hypothesis and conclude there is a significant difference in mean fuel consumption between the engine types.

Test assumptions

- We checked for normality (Shapiro-Wilk) and homogeneity of variances (Levene's test). Because the sample size is large (N > 5000), the normality test is overly sensitive, but the large sample size allows us to rely on the Central Limit Theorem.
- Variance was addressed by using the default robust t-test in R (Welch's t-test).

Construct 95% confidence intervals for the mean of two groups and interpret them

- We are 95% confident that the true population mean of FC for Piston engines lies within that interval **(29.61919, 30.62560)**.
- Similarly, the 95% CI for Turbofan is **(8.443816, 8.588554)**.

The two intervals **do not** overlap, this provides visual evidence that the mean Fuel Consumption differs significantly between the two engine types.

Check the association between Model and Sales Region in the whole sample using proper method and interpret your findings

```
table_model_region <- table(air_data$Model, air_data$SalesRegion)
chisq_result <- chisq.test(table_model_region)
print(chisq_result)
```

Conclusion

The p-value > 0.05 (~0.73), so we fail to reject the null hypothesis and conclude there is no statistically significant association between Model and Sales Region. The airplane models are distributed similarly across sales regions.

Filter your data only considering Bombardier CRJ200 and Cessna 172 model airplanes

```
filtered_air_data <- air_data |>
  filter(Model %in% c("Bombardier CRJ200", "Cessna 172")) |>
  droplevels()

str(filtered_air_data)
summary(filtered_air_data)
```

Check the distribution of Price across two categories of Engine type (You can apply transformation if you think it is required)

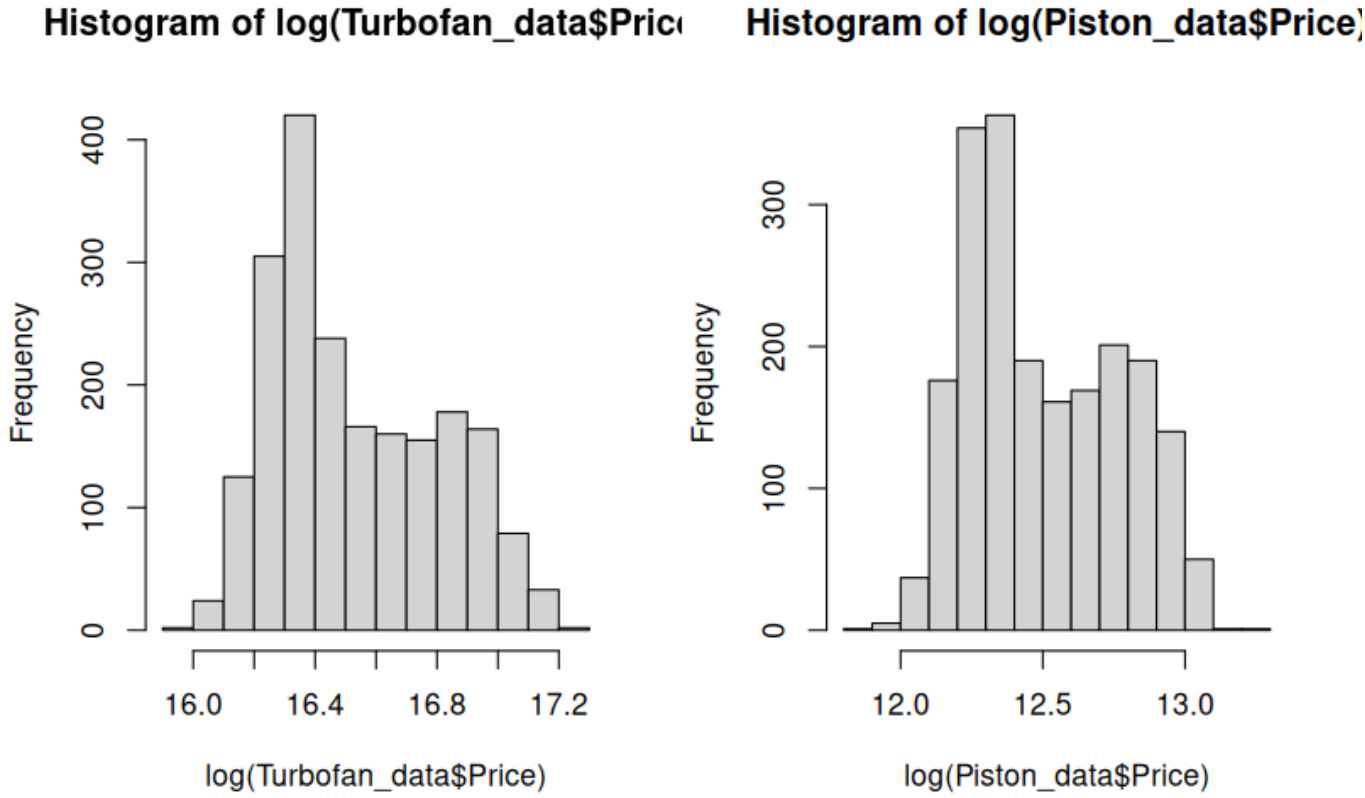

Figure 06

Both plots show a nearly normal distributed data across the two categories.

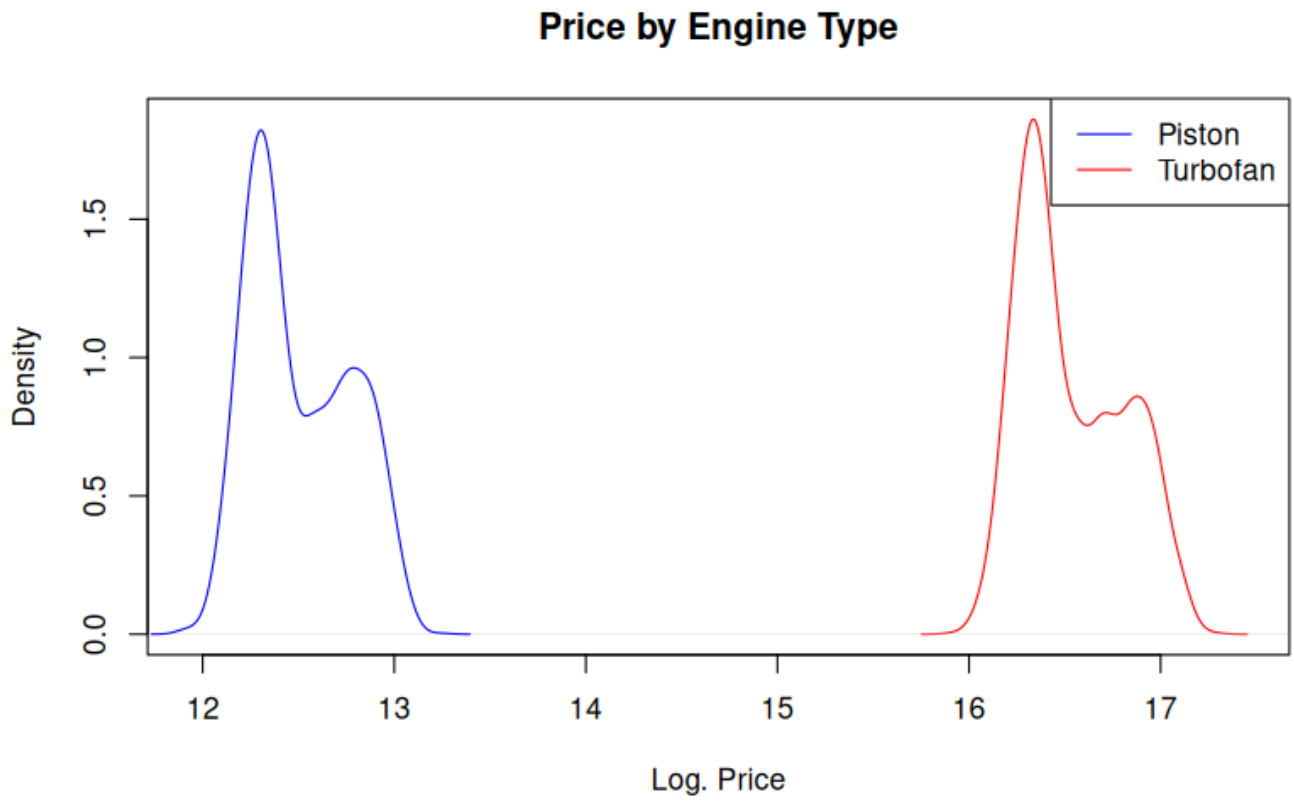

Figure 07

Conclusion

- We applied a log() transformation to Price to correct for positive (right) skewness and better approximate a normal distribution.
- We can conclude that there is a significant difference of Price between the two types of Engines.

Categorize the variable Price into two categories as “Low” and “High” by cutting from the median and save it as a new variable into your data frame

```
median_price <- median(filtered_air_data$Price, na.rm = TRUE)
filtered_air_data$Price_Category <- factor(
  ifelse(filtered_air_data$Price > median_price, "High", "Low")
)
```

Cross classify model and price categories and interpret the conditional probabilities

```
> # Conditional probabilities: P(PriceCategory | Model) - row proportions
> print(prop.table(cross_model_price, margin = 1))

                          High         Low
  Bombardier CRJ200 0.997074598 0.002925402
  Cessna 172        0.000000000 1.000000000
> # Conditional probabilities: P(Model | PriceCategory) - column proportions
> print(prop.table(cross_model_price, margin = 2))

                          High         Low
  Bombardier CRJ200 1.000000000 0.002933985
  Cessna 172        0.000000000 0.997066015
```

Figure 08

Interpretation of Conditional Probabilities

- **Row proportions P(PriceCategory | Model)**: show, for each model, what proportion of its airplanes fall into "Low" vs "High" price. Since Bombardier CRJ200 (Turbofan) has much higher prices than Cessna 172 (Piston), we expect nearly all Bombardier CRJ200 to be in the "High" category and nearly all Cessna 172 to be in the "Low" category.
- **Column proportions P(Model | PriceCategory)**: show, for each price category, what proportion belongs to each model. The "Low" category is dominated by Cessna 172 and the "High" category by Bombardier CRJ200. This also confirms a strong association between airplane model and price level.

Is there an association between the model of the airplane and its price level. Analyze it by using proper statistical method

```
test_model_price <- chisq.test(cross_model_price)
print(test_model_price)
```

Conclusion

The p-value < 0.05, so we reject the null hypothesis and conclude there is a statistically significant association between airplane model and price level, as expected.

Cross classify the variables Model and Sales region. Interpret the conditional probabilities and then test whether there is an association between these two characteristics

```
> # Conditional probabilities: P(SalesRegion | Model)
> print(prop.table(cross_model_region, margin = 1))

                      Africa      Asia Australia    Europe North America South America
  Bombardier CRJ200 0.1843003 0.1628474 0.1604096 0.1589469     0.1638225     0.1696733
  Cessna 172        0.1652771 0.1726336 0.1696910 0.1765571     0.1574301     0.1584110
> # Conditional probabilities: P(Model | SalesRegion)
> print(prop.table(cross_model_region, margin = 2))

                      Africa      Asia Australia    Europe North America South America
  Bombardier CRJ200 0.5286713 0.4868805 0.4874074 0.4752187     0.5114155     0.5186289
  Cessna 172        0.4713287 0.5131195 0.5125926 0.5247813     0.4885845     0.4813711
```

Figure 09

Interpretation of Conditional Probabilities

- **Row proportions P(SalesRegion | Model)**: shows the distribution of sales regions for each model. Both models have similar proportions across regions, so we conclude that the sales region is independent of the model.
- **Column proportions P(Model | SalesRegion)**: shows, within each region, what share belongs to each model. Both models appear in roughly equal proportions within every region, so this supports our independence assumption.

Conclusion

As expected, p-value > 0.05 (~0.29), so we fail to reject the null hypothesis and conclude there is no significant association. The two models are sold in similar proportions across regions.